



ATRIA INSTITUTE OF TECHNOLOGY

Project Title: Smart Car Parking System

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1. INTRODUCTION:

With the rapid growth of urban populations and vehicle usage, finding a suitable parking space has become a major challenge in cities. Traditional parking systems are often inefficient, time-consuming, and contribute to increased traffic congestion and pollution. To address these issues, Smart Car Parking Systems have emerged as an innovative solution that leverages technologies like Internet of Things (IoT), sensors, and automation.

A Smart Car Parking System is designed to automate the process of locating and managing parking spaces. It uses real-time data from sensors to detect available slots, displays information to users via apps or digital boards, and can even control gates for entry and exit. This system improves parking efficiency, reduces fuel consumption, saves time, and enhances the overall user experience.

2. ABSTRACT:

The increasing number of vehicles in urban areas has led to serious parking issues, including traffic congestion, fuel wastage, and increased air pollution. Traditional parking methods are often inefficient, time-consuming, and unable to provide real-time information about parking availability. To address these challenges, this project proposes a Smart Car Parking System using IoT (Internet of Things) technology.

The system utilizes sensors to detect the presence or absence of vehicles in parking slots and transmits this information in real-time to a central server or a mobile application via Wi-Fi. Users can view available parking spaces remotely through the app, enabling them to save time and reduce unnecessary driving. The system also includes automated gate control for secure entry and exit, enhancing convenience and safety.

This smart solution not only optimizes parking management but also contributes to reducing traffic congestion and environmental impact. It can be implemented in shopping malls, corporate offices, smart cities, and other public or private parking areas.

3. OBJECTIVES:

- To design and develop an automated car parking system.
- To detect real-time availability of parking slots.
- To provide users with live parking status.
- To minimize time spent searching for parking.
- To enhance security and control.
- To improve user convenience.
- To reduce manual intervention.

4. Components Used:

NodeMCU (ESP8266):- It is a low-cost, Wi-Fi-enabled microcontroller chip developed by Espressif Systems, widely used in IoT applications. It features a 32-bit processor, built-in Wi-Fi, GPIO pins, and support for communication protocols like UART, SPI, and I2C. The ESP8266 works by connecting to a Wi-Fi network and allowing microcontrollers or standalone programs to send and receive data over the internet. It receives instructions via serial communication or is programmed directly using environments like the Arduino IDE. The chip connects to servers, APIs, or cloud platforms, making it ideal for smart devices, data logging, remote monitoring, and automation projects.

Specifications:

The NodeMCU (ESP8266) is a compact, low-cost IoT development board based on the ESP8266EX microcontroller. It operates at 3.3V with a clock speed of 80 MHz (can go up to 160 MHz) and offers 11 digital I/O pins and 1 analog input. It features built-in Wi-Fi (802.11 b/g/n), supports protocols like UART, SPI, and I2C, and allows up to 5 simultaneous TCP connections. The board typically has 4 MB flash memory, 64 KB SRAM, and supports programming in Lua or C/C++ via the Arduino IDE. It includes a USB port for power/programming, a voltage regulator, onboard LED, and reset/flash buttons, making it ideal for IoT and embedded projects.

Features:

The NodeMCU (ESP8266) is a feature-rich, low-cost development board designed for IoT applications. It comes with built-in Wi-Fi support (802.11 b/g/n), allowing it to function as a station, access point, or both. Powered by the ESP8266EX microcontroller, it operates at 80 MHz (overclockable to 160 MHz) and includes 11 digital GPIO pins and one analog input. It supports popular communication protocols like UART, SPI, and I2C, and allows PWM on all digital pins. NodeMCU offers 4 MB of flash memory, a micro USB port for easy programming using CH340G or CP2102 USB-to-serial converters, and a built-in voltage regulator to convert 5V to 3.3V. It supports deep sleep mode for low power consumption, making it suitable for battery-powered projects. The board is programmable via Lua scripting or C/C++ using the Arduino IDE or PlatformIO. With its onboard LED, compact design, and open-source nature, NodeMCU is ideal for rapid IoT prototyping and embedded system development.

Node MCU ESP 8266

IR Sensor :- It is a device that detects infrared light emitted or reflected by objects. It works by emitting infrared light through an IR LED, which reflects back when it hits an object. The reflected light is then detected by a photodiode or phototransistor. If the reflected light is strong enough, the sensor activates and sends a digital or analog signal indicating the presence of an object. This non-contact detection method is widely used in applications such as obstacle detection, proximity sensing, and line-following systems.

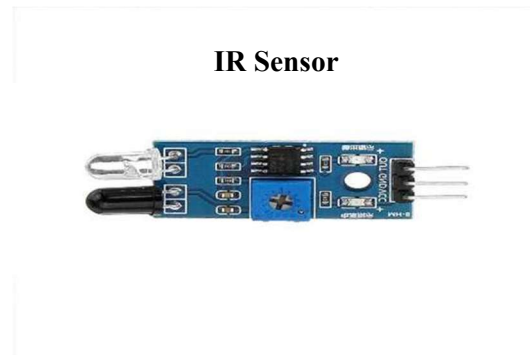
Specifications:

An IR (Infrared) sensor is an electronic device used primarily for detecting objects, measuring distance, or sensing proximity. It typically operates at a voltage range of 3V to 5.5V DC and consumes around 20 to 50 mA of current. The sensor consists of an IR LED as a transmitter and a photodiode or phototransistor as a receiver. It works within a wavelength range of 850 nm to 950 nm and can detect objects typically within a range of 2 cm to 80 cm, depending on surface reflectivity and ambient conditions. The output is usually digital (HIGH or LOW), but some models also provide analog signals. The IR sensor responds quickly, usually within 100

milliseconds, and has a viewing angle of around $\pm 15^\circ$ to $\pm 35^\circ$. Many IR sensors also include a built-in comparator and sensitivity adjustment via a potentiometer. Compact, cost-effective, and easy to interface with microcontrollers, IR sensors are commonly used in line-following robots, obstacle detection systems, and automation projects.

Features:

An IR (Infrared) sensor offers several useful features that make it ideal for object detection and proximity sensing. It enables non-contact detection, allowing it to sense obstacles without physical interaction. The sensor is compact, lightweight, and consumes very low power, making it suitable for portable and battery-operated devices. It delivers fast response times, typically under 100 milliseconds, and can provide either digital or analog output depending on the model. Many IR sensors come with adjustable sensitivity using a built-in potentiometer and feature onboard comparators like the LM358 for signal processing. They are easy to interface with microcontrollers such as Arduino and NodeMCU. While they perform well under normal lighting, their accuracy may decrease under strong ambient light. Overall, IR sensors are cost-effective, reliable, and widely used in robotics, automation, and electronic projects.



Servo Motor:- It is an electromechanical device designed for precise control of angular or linear position, commonly used in robotics, automation systems, and remote-controlled vehicles. It operates on low voltage (typically 4.8V to 6V) and is compact, lightweight, and capable of delivering high torque. The working principle of a servo motor is based on a closed-loop feedback system. It receives a control signal (usually PWM) that represents the desired position. Inside the motor, a potentiometer connected to the output shaft continuously monitors its position. The built-in control circuit compares this actual position with the target position and adjusts the motor's movement until they match. This feedback mechanism ensures accurate, smooth, and stable motion control, making servo motors highly reliable for precision applications.

Specifications:

A servo motor is a rotary actuator commonly used for precise control of angular position, typically in robotics, automation, and hobby electronics. It operates on 4.8V to 6V DC and is controlled using PWM (Pulse Width Modulation) signals. The motor can usually rotate between 0° and 180° , with some models allowing extended or continuous rotation. It includes a built-in control circuit and a feedback mechanism, often a potentiometer, to maintain accurate positioning. Servo motors like the SG90 offer a torque of around $1.8 \text{ kg}\cdot\text{cm}$ and are lightweight, while models like the MG996R provide higher torque ($\sim 10 \text{ kg}\cdot\text{cm}$) with metal gears for more demanding tasks. With three wires (power, ground, and signal), servo motors are easy to interface with microcontrollers, making them ideal for applications requiring controlled motion.

Features:

Servo motors offer precise control of angular position, making them ideal for applications like robotics, automation, and RC vehicles. They are compact, lightweight, and operate on low voltage (typically 4.8V to 6V). One of their key features is the built-in feedback system, usually using a potentiometer, which ensures accurate positioning. They are controlled by PWM signals, allowing smooth and adjustable movement within a defined range, usually 0° to 180°. Servo motors are also known for their high torque-to-size ratio, quick response time, and ease of interfacing with microcontrollers, making them highly reliable and efficient for motion control tasks.

Servo Motor

16x2 LCD I2C Display:- IT is a compact alphanumeric screen that shows 16 characters per line across 2 lines and communicates with a microcontroller using the I2C protocol, requiring only two wires (SDA and SCL). It includes an I2C adapter with a PCF8574 chip, which converts the serial data into parallel signals for the LCD's HD44780 controller. This setup simplifies wiring and saves I/O pins. The display works by lighting specific segments in a 5x8 dot matrix to form characters, with additional features like backlight and contrast control. It's ideal for space-saving and efficient embedded system projects.

Specifications:

It is an alphanumeric module that can show 16 characters per line on 2 lines, using a 5x8 dot matrix per character. It operates at 5V and communicates via the I2C protocol using only two pins: SDA and SCL, thanks to the onboard PCF8574 I/O expander. The display is controlled by an HD44780-compatible controller and typically has an I2C address of 0x27 or 0x3F. It features an LED backlight, an onboard contrast adjustment potentiometer, and is ideal for saving GPIO pins in microcontroller projects.

Features:

The offers several key features, making it ideal for embedded projects. It uses only 16x2 LCD I2C display two wires (SDA and SCL) for communication, saving microcontroller pins. The display can show 16 characters per line on 2 lines, with clear visibility provided by an LED backlight. It includes an onboard potentiometer for contrast adjustment and supports standard HD44780 commands. The PCF8574 I2C module enables easy interfacing, and the I2C address can be adjusted if needed. Its compact design, low power consumption, and ease of use make it a popular choice for displaying text in microcontroller-based applications.

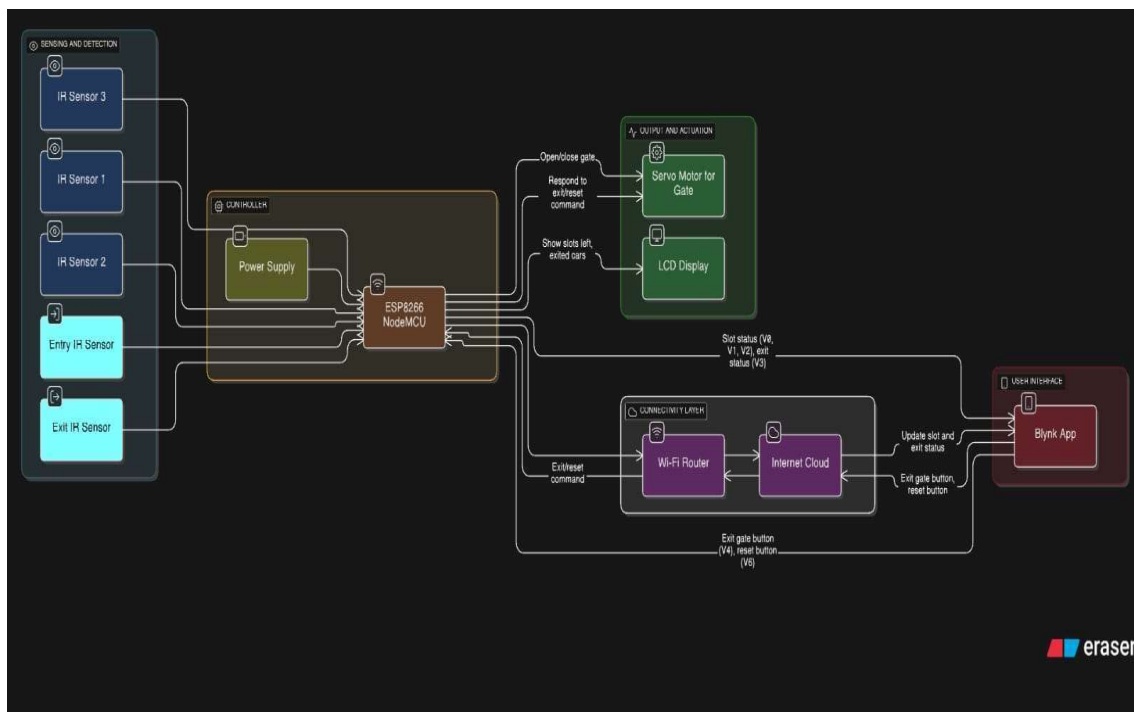
16*2 LCD Display I2C



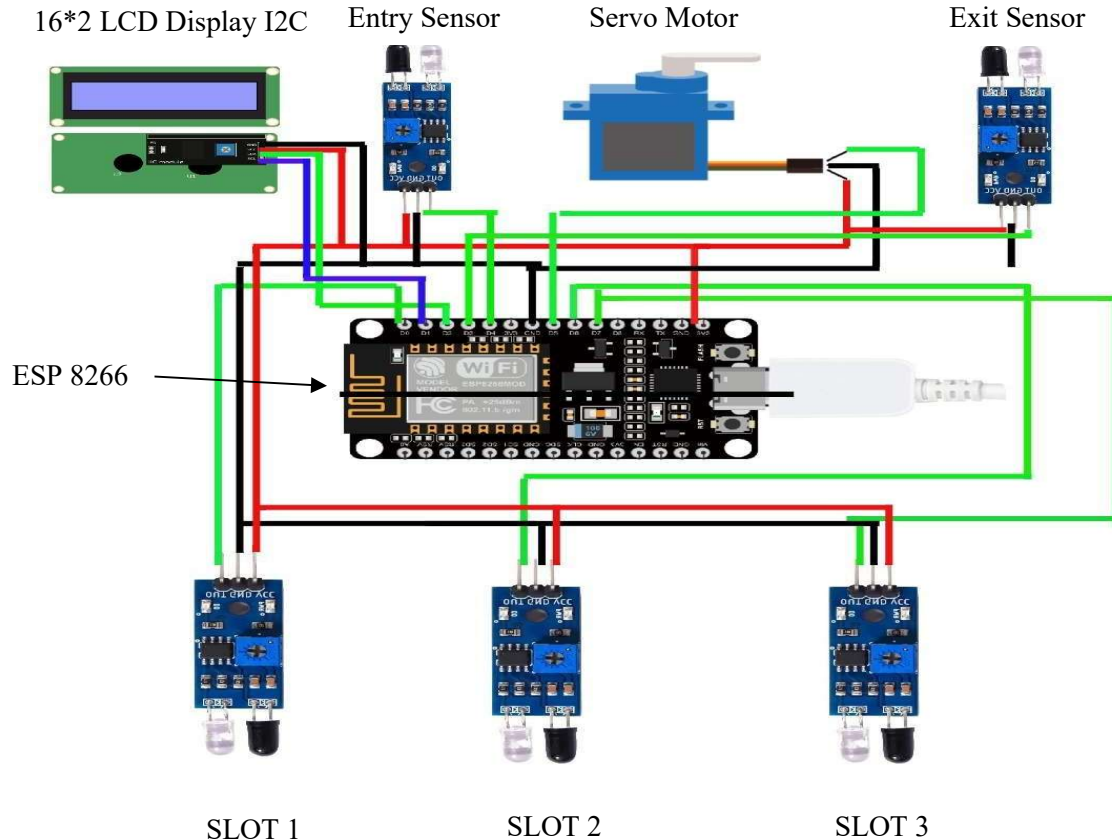
5. WORKING OF THE PROJECT:

The **working principle** of this smart parking system is based on the integration of IR sensors, a servo motor, an LCD display, and the Blynk IoT platform using a NodeMCU ESP8266. The system detects the presence of vehicles in parking slots using IR sensors and updates the availability status accordingly. When a vehicle approaches the entry point, an IR sensor detects it. If parking space is available, the servo motor opens the gate automatically and the available slot count is decreased. For vehicle exit, the gate is controlled manually using a button on the Blynk app; when pressed, the gate opens, the exit is counted, and the slot count increases. The current parking slot status, number of exited cars, and gate activities are displayed on a 16x2 I2C LCD and also sent to the Blynk app in real time. This system ensures efficient monitoring and management of parking space through both hardware automation and remote mobile access.

Block diagram:



CIRCUIT DIAGRAM:



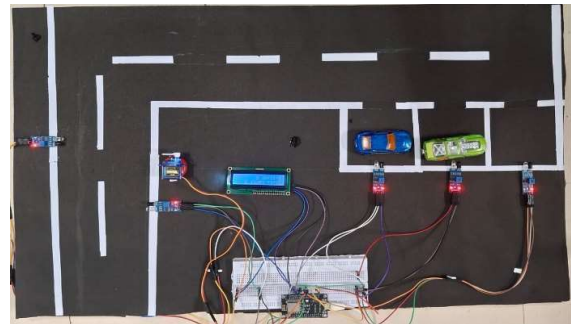
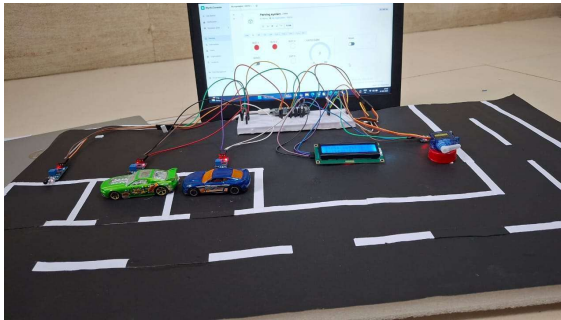
6. CLOUD CONNECTIVITY:

In this project, cloud connectivity is achieved through the Blynk platform, which allows the NodeMCU (ESP8266) to connect to the internet and communicate with the Blynk app using a unique authentication token. Once connected, the system can send real-time sensor data (like parking slot status and car exit count) to the cloud, which is then displayed on the mobile app. It also receives user commands from the app — such as opening the exit gate — via virtual pins. This enables remote monitoring and control of the parking system through the Blynk cloud without needing a local server.

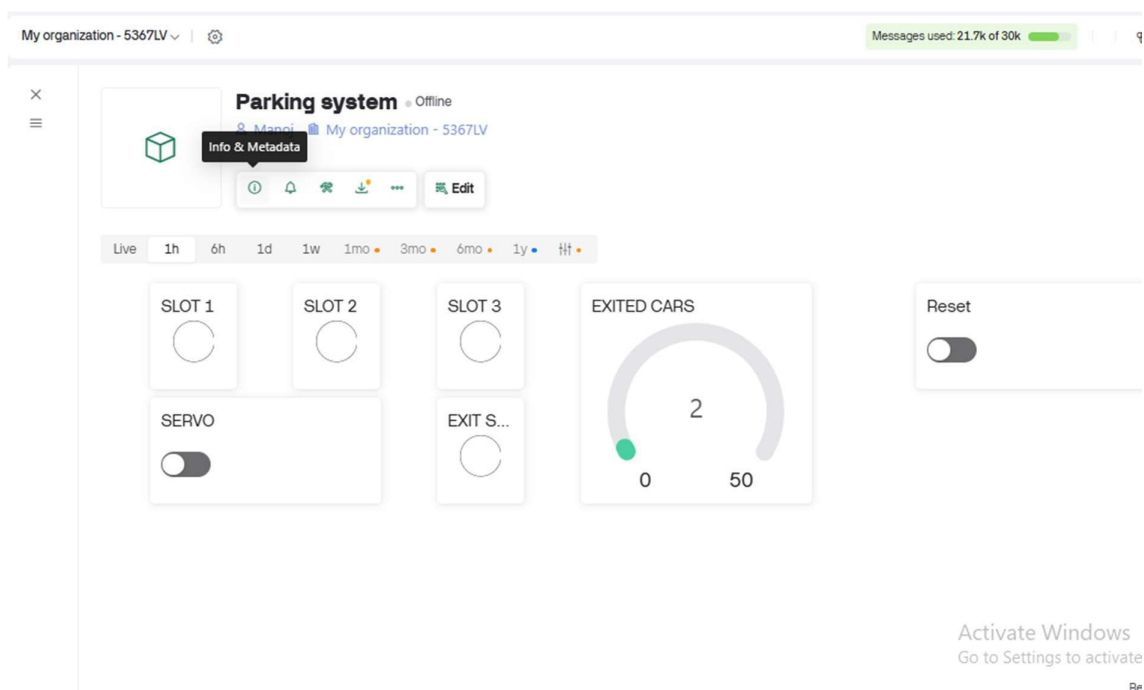
7. OUTPUT AND RESULTS:

The output of this parking system project is a real-time display and control of parking slot availability using sensors, an LCD, and the Blynk app. When a car arrives at the entry gate, the system detects it using an IR sensor. If a slot is available, the servo motor opens the gate, decreases the slot count, and updates the LCD and Blynk app accordingly. For exit, the user can manually trigger the gate using a virtual button in the Blynk app. When pressed, it increases the available slot count, increments the exit counter, and opens the gate. The LCD continuously displays the number of slots left and cars exited, while the Blynk app reflects the same data and also shows the real-time status of each parking slot using virtual LEDs. This provides a convenient and connected solution for smart parking management.

Results:



8. Blynk Dashboard Output:-



The Blynk IoT dashboard for the smart parking system project. It visually represents the status and control of a three-slot parking area. Each slot (SLOT 1, SLOT 2, SLOT 3) is monitored using IR sensors, and their availability is indicated through LED widgets. The number of cars that have exited the parking lot is shown on a gauge labeled "EXITED CARS," which is updated in real-time based on the code. There is a switch labeled "SERVO" that allows the user to manually open the exit gate via a servo motor when a car exits. Another LED indicator labeled "EXIT S..." reflects the status of the exit IR sensor, lighting up when a car is near the exit. Additionally, there is a "Reset" switch that, when programmed, can reset the parking data such as the exited car count and available slots. Overall, this dashboard allows users to remotely monitor and control the parking system using cloud connectivity through Blynk.

9. Future Scope:

This smart parking system can be expanded and improved in various ways for broader real-world applications. In the future, the system can integrate automatic payment gateways using

RFID, QR codes, or number plate recognition for seamless entry and billing. A mobile app interface can also be enhanced to include real-time parking availability across multiple parking lots within a city. Furthermore, integration with GPS and Google Maps can help users locate and reserve parking spots in advance. The system can also utilize machine learning to predict peak usage times and suggest optimal parking based on user history. Finally, solar-powered components and energy-efficient hardware can make the system more sustainable and suitable for smart cities.

10. Conclusion:

The smart parking system project successfully demonstrates an efficient and automated way to monitor and manage parking spaces using IoT technology. By integrating IR sensors, a servo motor, and a NodeMCU (ESP8266) with the Blynk IoT platform, the system can detect vehicle entry and exit, update slot availability, and display real-time data both on an LCD and the Blynk app. Cloud connectivity enables remote monitoring and control, allowing users to open the gate or reset the system through virtual switches. This project provides a practical solution to urban parking challenges by improving slot utilization, reducing manual effort, and offering remote access and control.

11. References

- NodeMCU Documentation
(<https://arduino-esp8266.readthedocs.io/en/latest/reference.html>)
- Blynk Docs
(<https://docs.blynk.io/en>)
- Sensor Datasheets (Servo Motor)
(<https://towerpro.com.tw/product/sg90-7/>)